Gibbons–Hawking–York Boundary Term
Boundary Term for General Relativity's Action in Vacuum

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ABSTRACT: The purpose of this essay is to show how to derive Einstein's equation in vacuum from an action, known as the *Einstein-Hilbert action*. In this framework, the metric is thought of as the dynamical variable. Consequently, we vary the action with respect to the metric while holding it fixed at the boundary. However, since the Lagrangian density is build upon the Ricci scalar, it inherently involves second derivatives of the metric. Therefore, the condition of fixed metric at the boundary alone is insufficient and we also need to demand that the variation of the derivatives vanishes on the boundary. Alternatively, we can incorporate the *Gibbons-Hawking-York Boundary Term* into the action, as we will explore further.

The convention we're going to use for the Riemann tensor in terms of the Christoffel symbols is

$$R^{\alpha}{}_{\mu\beta\nu} = \Gamma^{\alpha}_{\mu\nu,\beta} - \Gamma^{\alpha}_{\mu\beta,\nu} + \Gamma^{\lambda}_{\mu\nu}\Gamma^{\alpha}_{\lambda\beta} - \Gamma^{\lambda}_{\mu\beta}\Gamma^{\alpha}_{\lambda\nu},$$

where the Greek indices range from 0 to 3. Further, Latin indices range from 1 to 3, we adopt a metric with signature $(-1, 1, 1, 1)$, and we set $c = 1$. Unless otherwise specified, $(\mathcal{M}, \boldsymbol{g})$ denotes the four dimensional spacetime manifold with metric $\boldsymbol{g}$ and $g = \det(\boldsymbol{g})$.

KEYWORDS: General Relativity, Einstein–Hilbert Action, Boundary Terms, Gibbons–Hawking–York Term

Contents

Hypersurfaces and Gauss' Theorem

The purpose of this chapter is to introduce *non-null hypersurfaces* within $\mathcal{M}$ and their *induced metric*. Understanding these concepts is essential for grasping *Gauss' theorem*, which is crucial for integrating the action by parts. Additionally, we aim to introduce the notion of *extrinsic curvature* of hypersurfaces, which plays a crucial role in constructing the Gibbons-Hawking-York boundary term.

We take for granted that given any smooth oriented coordinates $\phi = \{x^i\}$ on $\mathcal{V} \subseteq \mathcal{M}$, the *metric volume form* has the local coordinate expression

$$\mathrm{d}V_g = \sqrt{-g}\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^4, \tag{1.0.1}$$

and can be used to integrate a function f over $\mathcal{V}$ by

$$\int_{\mathcal{V}} f\,\mathrm{d}V_g = \int_{\phi(\mathcal{V})} (\phi^{-1})^*(f\,\mathrm{d}V_g) = \int_{\phi(\mathcal{V})} f(x)\sqrt{-g}\,\mathrm{d}^4x. \tag{1.0.2}$$

The derivation of the previous formulas can be found in any standard textbooks like [1, Lee] or [2, Poisson]. Henceforth, we adopt a simplified notation, integrating over $\mathcal{V}$ for both the differential forms and their coordinate representation, without explicit reference to the chart ϕ.

1.1 Hypersurfaces on a Manifold

A *hypersurface* Σ is a three-dimensional submanifold embedded in $\mathcal{M}$, which can be either spacelike or timelike. We omit the consideration of the null case due to its increased technical complexity. The hypersurface can be described either by a scalar function $\Phi(x^\mu) = 0$ or by a parametric equation $x^\mu = x^\mu(y^a)$, where y^a are intrinsic coordinates of the hypersurface. The coordinate basis vectors related to y^a, and their dual forms, are given by[1]

$$e^\mu{}_a = \frac{\partial x^\mu}{\partial y^a}, \quad \tilde{e}_\mu{}^a = \frac{\partial y^a}{\partial x^\mu}, \tag{1.1.1}$$

with respect to the basis ∂_μ. They are related by $e^\mu{}_b \tilde{e}_\mu{}^a = \delta^a_b$.

Since $e^\mu{}_a$ form a basis for the tangent space to Σ, which is a subspace of the tangent space to $\mathcal{M}$, in order to construct a basis for the latter, we need to add the orthogonal complement to the former, by means of the *bulk metric* $g_{\mu\nu}$. This complement is spanned

[1] In particular, $\frac{\partial}{\partial y^a} = \frac{\partial x^\mu}{\partial y^a}\frac{\partial}{\partial x^\mu} = e^\mu{}_a \frac{\partial}{\partial x^\mu}$. Referring to "*vectors*", we understand their coordinates with respect to the coordinate basis of x^μ.

by a vector n^μ orthogonal to $e^\mu{}_a$. Assuming a normalization suitable for both spacelike and timelike hypersurfaces, we have

$$g_{\mu\nu} e^\mu{}_a n^\nu = 0, \quad g_{\mu\nu} n^\mu n^\nu = \epsilon, \tag{1.1.2}$$

where ϵ is defined as

$$\epsilon = n^\mu n_\mu \equiv \begin{cases} -1, & \text{if } \Sigma \text{ is spacelike,} \\ +1, & \text{if } \Sigma \text{ is timelike.} \end{cases} \tag{1.1.3}$$

Explicitly, the *unit normal vector* n^μ, pointing in the direction of increasing Φ, can be expressed as

$$n_\mu = \frac{\epsilon\, \partial_\mu \Phi}{\left| g^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi \right|^{\frac{1}{2}}}, \tag{1.1.4}$$

Note that, because of the normalization, the dual form associated to n^μ is not $n_\mu = g_{\mu\nu} n^\nu$, but rather $\tilde{n}_\mu = \epsilon n_\mu$. It satisfies

$$n^\mu \tilde{n}_\mu = 1, \quad n^\mu \tilde{e}_\mu{}^a = e^\mu{}_a \tilde{n}_\mu = 0, \quad n^\mu \tilde{n}_\nu + e^\mu{}_a \tilde{e}_\nu{}^a = \delta^\mu_\nu. \tag{1.1.5}$$

1.2 Projector onto the Hypersurface

The relations (1.1.5) lead us to define a *projector* $P^\mu{}_\nu$ as

$$P^\mu{}_\nu = \delta^\mu_\nu - n^\mu \tilde{n}_\nu = e^\mu{}_a \tilde{e}_\nu{}^a. \tag{1.2.1}$$

Using (1.1.2) and (1.2.1), we can show the following properties

$$P^\mu{}_\lambda P^\lambda{}_\nu = P^\mu{}_\nu, \tag{1.2.2a}$$
$$P^\mu{}_\nu e^\nu{}_a = e^\mu{}_a, \quad P^\mu{}_\nu n^\nu = 0, \tag{1.2.2b}$$
$$P^\mu{}_\nu \tilde{e}_\mu{}^a = \tilde{e}_\nu{}^a, \quad P^\mu{}_\nu \tilde{n}_\mu = 0, \tag{1.2.2c}$$

which define $P^\mu{}_\lambda$ as a projector of $\mathcal{M}$ onto the tangent space of Σ.

A vector field A^μ and a form A_μ are said to be *parallel* to the hypersurface if they can be written as

$$P^\mu{}_\nu A^\nu = A^\mu = A^a e^\mu{}_a \implies A^\mu n_\mu = 0, \tag{1.2.3a}$$
$$P^\nu{}_\mu A_\nu = A_\mu = A_a \tilde{e}_\mu{}^a \implies A_\mu n^\mu = 0, \tag{1.2.3b}$$

respectively. The generalization to generic tensors is straightforward.

1.3 Induced Metric

To define a metric on Σ, we can project the bulk metric $\boldsymbol{g}$ on it, using the projector (1.2.1). This gives us

$$h_{\mu\nu} = P^\alpha{}_\mu P^\beta{}_\nu g_{\alpha\beta} = h_{ab} \tilde{e}_\mu{}^a \tilde{e}_\nu{}^b, \tag{1.3.1a}$$
$$h_{ab} = e^\mu{}_a e^\nu{}_b h_{\mu\nu} = e^\mu{}_a e^\nu{}_b g_{\mu\nu}, \tag{1.3.1b}$$

which is called *induced metric* on Σ, and can be used to raise or lower submanifold's indices. In eq. (1.3.1a) we've written $\mathbf{h}$ as a parallel tensor, while the first equality in eq. (1.3.1b) is a consequence of the duality relation between $e^{\mu}{}_{a}$ and $\tilde{e}_{\mu}{}^{a}$ and the second one comes from (1.2.2).

By a substitution of (1.2.1) into (1.3.1a), one can show that the bulk metric $g_{\mu\nu}$ can be decomposed as

$$g_{\mu\nu} = h_{\mu\nu} + \epsilon n_{\mu} n_{\nu} \tag{1.3.2}$$

1.4 Gauss' Theorem

Using the induced metric on the hypersurface, we can introduce a surface element which allows us to integrate over Σ. Without delving too deeply into technical details, we present the results, which are proved in [2, Poisson].

Similar to (1.0.1), the invariant three-dimensional volume element on Σ, referred to as the *surface element* for simplicity, is given by

$$\mathrm{d}\Sigma \equiv \sqrt{|h|}\, \mathrm{d}^3 y, \tag{1.4.1}$$

where $h = \det(h_{ab})$. Furthermore, the *directed surface element*, which points in the direction of increasing Φ, is $n_{\alpha}\, \mathrm{d}\Sigma$, and it's useful to define

$$\mathrm{d}\Sigma_{\alpha} = \epsilon n_{\alpha}\, \mathrm{d}\Sigma. \tag{1.4.2}$$

Gauss' Theorem. *Let $\mathcal{V}$ denote a finite region of the spacetime manifold, bounded by a closed hypersurface $\partial\mathcal{V}$. For any vector field A^{α} defined within $\mathcal{V}$,*

$$\int_{\mathcal{V}} \nabla_{\alpha} A^{\alpha} \sqrt{-g}\, \mathrm{d}^4 x = \oint_{\partial\mathcal{V}} A^{\alpha}\, \mathrm{d}\Sigma_{\alpha}, \tag{1.4.3}$$

where $\mathrm{d}\Sigma_{\alpha}$ is the surface element defined by (1.4.2).

1.5 Induced Covariant Derivatives and Extrinsic Curvature

Let's denote with ∇ the covariant derivative defined with respect to $\mathbf{g}$. Taking two vectors of Σ, the covariant derivative of a vector in the direction of the other is not, in general, a vector of Σ. Therefore, it's natural to define a covariant derivative in Σ by requiring it to be a vector of the hypersurface.

Let's then consider a vector A^{μ} and a form A_{μ} parallel to the hypersurface. Their *induced covariant derivative* on Σ is defined by

$$\begin{aligned} D_{\alpha} A_{\beta} &\equiv P^{\mu}{}_{\alpha} P^{\nu}{}_{\beta} \nabla_{\mu} A_{\nu} = D_a A_b \tilde{e}_{\alpha}{}^{a} \tilde{e}_{\beta}{}^{b}, \quad D_a A_b = \nabla_{\mu} A_{\nu} e^{\mu}{}_{a} e^{\nu}{}_{b}, \\ D_{\alpha} A^{\beta} &\equiv P^{\mu}{}_{\alpha} P^{\beta}{}_{\nu} \nabla_{\mu} A^{\nu} = D_a A^b \tilde{e}_{\alpha}{}^{a} e^{\beta}{}_{b}, \quad D_a A^b = \nabla_{\mu} A^{\nu} e^{\mu}{}_{a} \tilde{e}_{\nu}{}^{b}. \end{aligned} \tag{1.5.1}$$

Since they are parallel, the condition $n_{\nu} A^{\nu} = 0$ holds, and this allows us to write $n_{\nu} \nabla_{\mu} A^{\nu} = -A^{\nu} \nabla_{\mu} n_{\nu}$. Using the latter expression, we can compute

$$\begin{aligned} D_{\alpha} A^{\beta} &= P^{\mu}{}_{\alpha} P^{\beta}{}_{\nu} \nabla_{\mu} A^{\nu} = P^{\mu}{}_{\alpha} (\delta^{\beta}_{\nu} - \epsilon n^{\beta} n_{\nu}) \nabla_{\mu} A^{\nu} = P^{\mu}{}_{\alpha} \nabla_{\mu} A^{\beta} - \epsilon P^{\mu}{}_{\alpha} n^{\beta} n_{\nu} \nabla_{\mu} A^{\nu} \\ &= P^{\mu}{}_{\alpha} \nabla_{\mu} A^{\beta} + \epsilon P^{\mu}{}_{\alpha} n^{\beta} A^{\nu} \nabla_{\mu} n_{\nu} \equiv D_a A^b \tilde{e}_{\alpha}{}^{a} e^{\beta}{}_{b}. \end{aligned}$$

Then, multiplying both sides of the last line by $e^{\alpha}{}_{c}$, using (1.2.2) and relabelling $(c \to a)$, yields

$$D_a A^b e^{\beta}{}_b = \nabla_\mu A^\beta e^\mu{}_a + \epsilon n^\beta e^\mu{}_a A^\nu \nabla_\mu n_\nu = \nabla_\mu A^\beta e^\mu{}_a + \epsilon n^\beta e^\mu{}_a A^b e^\nu{}_b \nabla_\mu n_\nu .$$

Now, defining the *extrinsic curvature* as

$$K_{ab} \equiv \nabla_\mu n_\nu e^\mu{}_a e^\nu{}_b, \tag{1.5.2}$$

leads to

$$\nabla_\mu A^\beta e^\mu{}_a = D_a A^b e^\beta{}_b - \epsilon A^b K_{ab} n^\beta. \tag{1.5.3}$$

Then, the intrinsic covariant derivative gives us a tangential component of the covariant derivative, while $-\epsilon A^a K_{ab}$ represents a normal component. One can further prove that the extrinsic curvature is a symmetric tensor. Its trace is given by

$$K \equiv h^{ab} K_{ab} = \nabla_\mu n^\mu, \tag{1.5.4}$$

where we've used eq. (1.3.1) to write $h^{ab} e^\mu{}_a e^\nu{}_b = g^{\mu\nu}$.

As we shall see, the Gibbons-Hawking-York boundary term is constructed out of the trace of the extrinsic curvature.

Einstein–Hilbert Action

In this chapter, we aim to formulate the *Einstein–Hilbert action* and derive *Einstein's equations* from it. As discussed in the introduction, the metric provides a natural volume form (1.0.1). By multiplying it by a scalar function, we can construct an integral similar to (1.0.2). Considering the metric $g_{\mu\nu}$ as the dynamical variable, the simplest scalar constructable from the metric, which is no higher than second order in its derivatives, is the *Ricci scalar* R. This leads us to the following action

$$S_{EH} = \int_{\mathcal{M}} \mathrm{d}^4x \sqrt{-g} R, \tag{2.0.1}$$

2.1 Variation of the action

As usual, let's compute the variation of the action with respect to the metric and set it to zero to obtain an extremum, which represents the physical equations of motion,

$$\delta S_{EH} = \int_{\mathcal{M}} \mathrm{d}^4x \Big[\underbrace{(\delta\sqrt{-g})g^{\mu\nu}R_{\mu\nu}}_{\delta S_1} + \underbrace{\sqrt{-g}(\delta g^{\mu\nu})R_{\mu\nu}}_{\delta S_2} + \underbrace{\sqrt{-g}\, g^{\mu\nu}\delta R_{\mu\nu}}_{\delta S_3}\Big]. \tag{2.1.1}$$

It's convenient to perform the variation with respect to the inverse metric $g^{\mu\nu}$, defined by $g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu$. Since the Kronecker delta doesn't change under a variation of the metric, it's easy to compute $(\delta g^{\rho\mu})g_{\mu\nu} + g^{\rho\mu}(\delta g_{\mu\nu}) = 0 \implies \delta g_{\sigma\nu} = -g_{\sigma\rho}g_{\mu\nu}\delta g^{\rho\mu}$, where we've multiplied both sides by $g_{\sigma\rho}$. Relabelling and using the symmetry of the metric leads to

$$\delta g_{\mu\nu} = -g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}. \tag{2.1.2}$$

This allows us to compute the variation of the determinant as

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}. \tag{2.1.3}$$

Proof. Let's first consider any diagonalizable matrix $\boldsymbol{M}$ and compute the variation of $\ln|\det \boldsymbol{M}|$ induced by a variation of $\boldsymbol{M}$. Using that the product of determinants is the determinant of the product and the identity $\ln\det\boldsymbol{M} = \operatorname{tr}\ln\boldsymbol{M}$,

$$\begin{aligned}
\delta\ln|\det\boldsymbol{M}| &\equiv \ln|\det(\boldsymbol{M}+\delta\boldsymbol{M})| - \ln|\det\boldsymbol{M}| = \ln\frac{\det(\boldsymbol{M}+\delta\boldsymbol{M})}{\det\boldsymbol{M}} \\
&= \ln\det\boldsymbol{M}^{-1}(\boldsymbol{M}+\delta\boldsymbol{M}) = \ln\det(\boldsymbol{1}+\boldsymbol{M}^{-1}\delta\boldsymbol{M}) \\
&= \operatorname{tr}\ln(\boldsymbol{1}+\boldsymbol{M}^{-1}\delta\boldsymbol{M}) \simeq \operatorname{tr}\boldsymbol{M}^{-1}\delta\boldsymbol{M},
\end{aligned}$$

where the last equality holds for small variations. If we replace $\boldsymbol{M}$ with the metric tensor $\boldsymbol{g}$, we have

$$\frac{1}{\det \boldsymbol{g}}\delta \det \boldsymbol{g} = \frac{1}{g}\delta g = \mathrm{tr}(\boldsymbol{g}^{-1}\delta \boldsymbol{g}) = g^{\mu\nu}\delta g_{\mu\nu}.$$

Finally, making use of the last expression and eq. (2.1.2) leads to

$$\begin{aligned}\delta\sqrt{-g} &= -\frac{1}{2\sqrt{-g}}\delta g = -\frac{1}{2}\frac{g}{\sqrt{-g}}g^{\mu\nu}\delta g_{\mu\nu} = \frac{1}{2}\frac{g}{\sqrt{-g}}g^{\mu\nu}g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma} \\ &= \frac{1}{2}\frac{g}{\sqrt{-g}}g_{\rho\sigma}\delta g^{\rho\sigma} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu},\end{aligned}$$

as claimed. ■

Therefore, eq. (2.1.1) can be written as

$$\delta S_{EH} = \int_{\mathcal{M}} \mathrm{d}^4x\sqrt{-g}\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right)\delta g^{\mu\nu} + \int_{\mathcal{M}} \mathrm{d}^4x\,\sqrt{-g}\,g^{\mu\nu}\delta R_{\mu\nu}, \tag{2.1.4}$$

which shows us that to recover Einstein's equations in vacuum, the last integral must not contribute. Indeed, it turns out to be a total derivative.

To show it, let's first compute the variation of the Christoffel symbols. Even if they don't form a tensor, their variation do. Indeed, it can be expressed as

$$\delta\Gamma^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\nabla_\nu\delta g_{\mu\sigma} + \nabla_\mu\delta g_{\nu\sigma} - \nabla_\sigma\delta g_{\mu\nu}). \tag{2.1.5}$$

Proof. Let's start with the expression of the Christoffel symbols in terms of the metric, i.e.,

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\nu g_{\mu\sigma} + \partial_\mu g_{\nu\sigma} - \partial_\sigma g_{\mu\nu}). \tag{2.1.6}$$

Varying it,

$$\begin{aligned}\delta\Gamma^{\rho}_{\mu\nu} =& \ \frac{1}{2}\delta g^{\rho\sigma}(\partial_\nu g_{\mu\sigma} + \partial_\mu g_{\nu\sigma} - \partial_\sigma g_{\mu\nu}) + \frac{1}{2}g^{\rho\sigma}(\partial_\nu\delta g_{\mu\sigma} + \partial_\mu\delta g_{\nu\sigma} - \partial_\sigma\delta g_{\mu\nu}) \\ =& -\frac{1}{2}g^{\rho\alpha}g^{\sigma\beta}\delta g_{\alpha\beta}(\partial_\nu g_{\mu\sigma} + \partial_\mu g_{\nu\sigma} - \partial_\sigma g_{\mu\nu}) + \frac{1}{2}g^{\rho\sigma}(\partial_\nu\delta g_{\mu\sigma} + \partial_\mu\delta g_{\nu\sigma} - \partial_\sigma\delta g_{\mu\nu}) \\ =& -g^{\rho\alpha}\delta g_{\alpha\beta}\Gamma^{\beta}_{\mu\nu} + \frac{1}{2}g^{\rho\sigma}(\partial_\nu\delta g_{\mu\sigma} + \partial_\mu\delta g_{\nu\sigma} - \partial_\sigma\delta g_{\mu\nu}) \\ =& \ \frac{1}{2}g^{\rho\sigma}(\partial_\nu\delta g_{\mu\sigma} + \partial_\mu\delta g_{\nu\sigma} - \partial_\sigma\delta g_{\mu\nu} - 2\delta g_{\sigma\lambda}\Gamma^{\lambda}_{\mu\nu}) \\ =& \ \frac{1}{2}g^{\rho\sigma}(\partial_\nu\delta g_{\mu\sigma} - \Gamma^{\lambda}_{\sigma\nu}\delta g_{\mu\lambda} - \Gamma^{\lambda}_{\mu\nu}\delta g_{\lambda\sigma} + \partial_\mu\delta g_{\nu\sigma} - \Gamma^{\lambda}_{\sigma\mu}\delta g_{\nu\lambda} - \Gamma^{\lambda}_{\nu\mu}\delta g_{\lambda\sigma} \\ & \quad - \partial_\sigma\delta g_{\mu\nu} + \Gamma^{\lambda}_{\mu\sigma}\delta g_{\lambda\nu} + \Gamma^{\lambda}_{\nu\sigma}\delta g_{\mu\lambda}) \\ =& \ \frac{1}{2}g^{\rho\sigma}(\nabla_\nu\delta g_{\mu\sigma} + \nabla_\mu\delta g_{\nu\sigma} - \nabla_\sigma\delta g_{\mu\nu})\end{aligned}$$

where in the second line we've used eq. (2.1.2), in the third eq. (2.1.6), in the fourth we've relabelled ($\alpha \to \sigma$) and ($\beta \to \lambda$) and in the fifth we've added and subtracted two terms, using both the symmetry of the metric and the Christoffel symbols. This completes the proof. ■

It's now straightforward to show that the variation of the Ricci tensor is given by

$$\delta R_{\mu\nu} = \nabla_\lambda \delta \Gamma^\lambda_{\mu\nu} - \nabla_\nu \delta \Gamma^\lambda_{\mu\lambda} \tag{2.1.7}$$

Proof. First, since by (2.1.5) the variation of the Christoffel symbols is a tensor, we can compute its covariant derivative, namely

$$\nabla_\sigma \delta \Gamma^\rho_{\mu\nu} = \partial_\sigma \delta \Gamma^\rho_{\mu\nu} + \Gamma^\rho_{\lambda\sigma} \delta \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\sigma} \delta \Gamma^\rho_{\mu\lambda} - \Gamma^\lambda_{\mu\sigma} \delta \Gamma^\rho_{\lambda\nu}. \tag{2.1.8}$$

Then, using the expression of the Riemann tensor in terms of the Christoffel symbols,

$$R^\rho{}_{\mu\sigma\nu} = \partial_\sigma \Gamma^\rho_{\mu\nu} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\lambda_{\mu\nu} \Gamma^\rho_{\lambda\sigma} - \Gamma^\lambda_{\mu\sigma} \Gamma^\rho_{\lambda\nu}, \tag{2.1.9}$$

and using (2.1.8) to rewrite the partial derivatives in the previous expression, leads to

$$\begin{aligned}
\delta R^\rho{}_{\mu\sigma\nu} &= \partial_\sigma \delta \Gamma^\rho_{\mu\nu} - \partial_\nu \delta \Gamma^\rho_{\mu\sigma} + \delta \Gamma^\lambda_{\mu\nu} \Gamma^\rho_{\lambda\sigma} + \Gamma^\lambda_{\mu\nu} \delta \Gamma^\rho_{\lambda\sigma} - \delta \Gamma^\lambda_{\mu\sigma} \Gamma^\rho_{\lambda\nu} - \Gamma^\lambda_{\mu\sigma} \delta \Gamma^\rho_{\lambda\nu} \\
&= \nabla_\sigma \delta \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\lambda\sigma} \delta \Gamma^\lambda_{\mu\nu} + \Gamma^\lambda_{\nu\sigma} \delta \Gamma^\rho_{\mu\lambda} + \Gamma^\lambda_{\mu\sigma} \delta \Gamma^\rho_{\lambda\nu} \\
&\quad - \nabla_\nu \delta \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\lambda\nu} \delta \Gamma^\lambda_{\mu\sigma} - \Gamma^\lambda_{\sigma\nu} \delta \Gamma^\rho_{\mu\lambda} - \Gamma^\lambda_{\mu\nu} \delta \Gamma^\rho_{\lambda\sigma} \\
&\quad + \Gamma^\rho_{\lambda\sigma} \delta \Gamma^\lambda_{\mu\nu} + \Gamma^\lambda_{\mu\nu} \delta \Gamma^\rho_{\lambda\sigma} - \Gamma^\rho_{\lambda\nu} \delta \Gamma^\lambda_{\mu\sigma} - \Gamma^\lambda_{\mu\sigma} \delta \Gamma^\rho_{\lambda\nu} \\
&= \nabla_\sigma \delta \Gamma^\rho_{\mu\nu} - \nabla_\nu \delta \Gamma^\rho_{\mu\sigma}.
\end{aligned}$$

Now, contracting the first and the third indices, and denoting them by λ, namely, $\delta R^\lambda_{\mu\lambda\nu} \equiv \delta R_{\mu\nu}$, completes the proof. ■

Now, let's rewrite the term δS_3 of (2.1.1) as a total derivative, and apply Gauss' theorem. Using the result (2.1.7), the metric compatibility condition $\nabla \boldsymbol{g} = 0$ and relabelling a couple of indices, we obtain

$$\begin{aligned}
\delta S_3 &= \int_{\mathcal{M}} \mathrm{d}^4 x \sqrt{-g}\, g^{\mu\nu} \delta R_{\mu\nu} = \int_{\mathcal{M}} \mathrm{d}^4 x \sqrt{-g}\, g^{\mu\nu} (\nabla_\lambda \delta \Gamma^\lambda_{\mu\nu} - \nabla_\nu \delta \Gamma^\lambda_{\mu\lambda}) \\
&= \int_{\mathcal{M}} \mathrm{d}^4 x \sqrt{-g}\, \nabla_\alpha (g^{\mu\nu} \delta \Gamma^\alpha_{\mu\nu} - g^{\mu\alpha} \delta \Gamma^\lambda_{\mu\lambda}) \equiv \int_{\mathcal{M}} \mathrm{d}^4 x \sqrt{-g}\, \nabla_\alpha V^\alpha,
\end{aligned} \tag{2.1.10}$$

where we've defined

$$V^\alpha = g^{\mu\nu} \delta \Gamma^\alpha_{\mu\nu} - g^{\mu\alpha} \delta \Gamma^\lambda_{\mu\lambda}. \tag{2.1.11}$$

We can now apply Gauss' theorem (1.4.3), so that the variation of the action, given by (2.1.4), becomes

$$\delta S_{EH} = \int_{\mathcal{M}} \mathrm{d}^4 x \sqrt{-g}\, G_{\mu\nu} \delta g^{\mu\nu} + \oint_{\partial\mathcal{M}} \mathrm{d}^3 y\, \epsilon \sqrt{|h|}\, n_\alpha V^\alpha. \tag{2.1.12}$$

Setting the boundary term to zero leads to Einstein's equation in vacuum, i.e., $G^{\mu\nu} = 0$. However, the matter turns out to be subtler.

2.2 Gibbons-Hawking-York boundary term

Examining the definition of V^α, eq. (2.1.11) and consequently the one of $\delta\Gamma^\rho_{\mu\nu}$, eq. (2.1.5), we observe that the surface term contains both the variation of the metric, $\delta g_{\mu\nu}$, and of its derivatives, $\delta(\partial_\alpha g_{\mu\nu})$. Therefore, simply setting $\delta g_{\mu\nu} = 0$ at the boundary is not sufficient to eliminate all the surface contributions of (2.1.12). For this reason Gibbons, Hawking, and York proposed to add the term

$$S_{GHY} = 2\oint_{\partial\mathcal{M}} \mathrm{d}^3 y\, \epsilon\sqrt{|h|}\, K, \tag{2.2.1}$$

where K is the trace of the extrinsic curvature of the boundary. By doing so, the variation δS_{GHY} cancels the term involving $\delta(\partial_\alpha g_{\mu\nu})$, making it sufficient to set $\delta g_{\mu\nu} = 0$ at the boundary to render the action stationary. To understand how this is accomplished, it's convenient to rewrite the surface contribution of (2.1.12) in a more suitable form.

To compute the variation of the action with respect to the metric, we assume that on the boundary $\partial\mathcal{M}$ the metric is held fixed, i.e., $\delta g_{\mu\nu}\big|_{\partial\mathcal{M}} = 0 = \delta g^{\mu\nu}\big|_{\partial\mathcal{M}}$. This allows us to compute, starting from (2.1.5),

$$\delta\Gamma^\rho_{\mu\nu}|_{\partial\mathcal{M}} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu\delta g_{\sigma\nu} + \partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}), \tag{2.2.2}$$

so that, at the boundary,

$$\begin{aligned}
V^\alpha &= g^{\mu\nu}\delta\Gamma^\alpha_{\mu\nu} - g^{\mu\alpha}\delta\Gamma^\lambda_{\mu\lambda} \\
&= \frac{1}{2}g^{\mu\nu}g^{\alpha\sigma}(\partial_\mu\delta g_{\sigma\nu} + \partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}) - \frac{1}{2}g^{\mu\alpha}g^{\lambda\sigma}(\partial_\mu\delta g_{\sigma\lambda} + \partial_\lambda\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\lambda}) \\
&= \frac{1}{2}g^{\mu\nu}g^{\alpha\sigma}(\partial_\mu\delta g_{\sigma\nu} + \partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}) - \frac{1}{2}g^{\sigma\alpha}g^{\mu\nu}(\partial_\sigma\delta g_{\nu\mu} + \partial_\mu\delta g_{\nu\sigma} - \partial_\nu\delta g_{\sigma\mu}) \\
&= g^{\mu\nu}g^{\alpha\sigma}(\partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}),
\end{aligned}$$

where, for the second part of the third line we relabelled $(\mu\to\sigma)$,$(\lambda\to\mu)$ and $(\sigma\to\nu)$. Then,

$$\begin{aligned}
n_\alpha V^\alpha|_{\partial\mathcal{M}} &= n_\alpha g^{\mu\nu}g^{\alpha\sigma}(\partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}) = n^\sigma(\epsilon n^\mu n^\nu + h^{\mu\nu})(\partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}) \\
&= n^\sigma h^{\mu\nu}(\partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}),
\end{aligned} \tag{2.2.3}$$

where in the first line we've used the decomposition (1.3.2) and in the second line we've used that $n^\sigma n^\nu$ is symmetric in (σ-ν) while the quantity within the brackets is antisymmetric with respect to those indices.

Further, since $\delta g_{\mu\nu}$ vanishes everywhere on the boundary $\partial\mathcal{M}$, its tangential derivatives must also vanish, namely

$$\delta g_{\mu\nu}\big|_{\partial\mathcal{M}} = 0 \implies \partial_\alpha\delta g_{\mu\nu}e^\alpha{}_a\big|_{\partial\mathcal{M}} = 0 \tag{2.2.4}$$

Therefore,

$$h^{\mu\nu}\partial_\nu\delta g_{\sigma\mu} = h^{ab}e^\mu{}_a e^\nu{}_b\partial_\nu\delta g_{\sigma\mu} = 0,$$

and so

$$n_\alpha V^\alpha|_{\partial\mathcal{M}} = -n^\sigma h^{\mu\nu}\partial_\sigma\delta g_{\mu\nu}. \tag{2.2.5}$$

Combining everything, the surface term of (2.1.12) is

$$\delta S_3 = -\oint_{\partial\mathcal{M}} \mathrm{d}^3 y\, \epsilon \sqrt{|h|}\, h^{\mu\nu} \partial_\alpha \delta g_{\mu\nu} n^\alpha. \tag{2.2.6}$$

Let's now vary the Gibbons, Hawking, and York boundary term (2.2.1) and show that it cancels out the previous surface term. To do so, let's first notice that by eq. (1.3.1b), if the variation of the bulk metric is zero at the boundary, the induced metric is held fixed too. Therefore, the only term to vary is K, so that

$$\delta S_{GHY} = 2 \oint_{\partial\mathcal{M}} \mathrm{d}^3 y\, \epsilon \sqrt{|h|}\, \delta K. \tag{2.2.7}$$

To compute its variation, it's convenient to write the trace of the extrinsic curvature, defined by (1.5.4), as

$$K = \nabla_\mu n^\mu = g^{\mu\nu} \nabla_\mu n_\nu = (h^{\mu\nu} + \epsilon n^\mu n^\nu) \nabla_\mu n_\nu = h^{\mu\nu} (\partial_\mu n_\nu - \Gamma^\lambda_{\mu\nu} n_\lambda), \tag{2.2.8}$$

where we've used the metric compatibility of the covariant derivative, the decomposition (1.3.2), the fact that $\nabla_\mu (n_\nu n^\nu) = 0$ implies $n^\nu \nabla_\mu n_\nu = 0$, and the definition of the covariant derivative. Then, using (2.2.2),we can compute

$$\delta K = -h^{\mu\nu} \delta\Gamma^\lambda_{\mu\nu} n_\lambda = -\frac{1}{2} h^{\mu\nu} g^{\lambda\sigma} n_\lambda (\partial_\mu \delta g_{\sigma\nu} + \partial_\nu \delta g_{\sigma\mu} - \partial_\sigma \delta g_{\mu\nu}) = \frac{1}{2} h^{\mu\nu} n^\sigma \partial_\sigma \delta g_{\mu\nu},$$

where we've again used that the induced metric is held fixed at the boundary while the tangential derivatives of $\delta g_{\mu\nu}$ vanish there. Therefore, we have

$$\delta S_{GHY} = \oint_{\partial\mathcal{M}} \mathrm{d}^3 y\, \epsilon \sqrt{|h|}\, h^{\mu\nu} \partial_\alpha \delta g_{\mu\nu} n^\alpha, \tag{2.2.9}$$

which cancels out (2.2.6).

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